MGT 404 Problem Set 2

NOTE: YOU MUST SUBMIT YOUR ANSWERS VIA CANVAS. NO HARD-COPY SUBMISSIONS WILL BE GRADED.

1. *Aluminum.* Aluminum is produced by adding large amounts of electricity to aluminum oxide. A manager of a smelting plant considering entering the aluminum industry estimates that combining e thousand kilowatt-hours of electricity and a metric tons of aluminum oxide creates:

$$q = f(e, a) = 10(9 * e * a)^{1/4}$$

metric tons of aluminum metal.

a) The price of electricity is \$50 per thousand kilowatt-hours and the price of aluminum oxide is currently \$200 per metric ton. Solve for the MP of e and a.

MP_e=______________________ MP_a=______________________

b) In a cost-minimizing production plan, how many thousand kilowatt-hours of electricity should be used for every metric ton of aluminum oxide?

KW Hours (000's) = ____________________________________

c) Suppose the smelting plant manager wants to produce 3,000 tons of aluminum. How many thousand kilowatt-hours of electricity and how many tons of aluminum oxide should be used?

KW Hours of Electricity (000's) = ________________________

Tons of Aluminum Oxide = ___________________________

d) Once the manager enters the aluminum industry, she realizes that a smelting plant producing q metric tons of aluminum actually has a daily variable cost of:

$$VC(q) = 0.5q^2$$

Throughout, let p denote the price per metric ton of aluminum. Note that you do not need to use the production function from (a) to answer the following.

What is the plant's short-run supply curve, q(p)? Also, assuming there are 100 such plants (with an identical cost structure) in the world, what is the global supply of aluminum $Q_S^G(p)$?

Short-Run Supply, q(p): ____________________

Global Supply, $Q_S^G(p)$?: ____________________

e) Daily demand for aluminum in the global market is:

$$P = \frac{8{,}000}{3} - \frac{1}{300} Q_D^G(P)$$

What is the equilibrium price and daily quantity of aluminum sold per plant?

Equilibrium Price: ____________

Quantity Sold Per Plant: ____________

f) Now consider the entry of Iceland into the global aluminum market. Suppose that there are 10 aluminum producers in Iceland that can produce relatively cheaply using geothermal energy, with:

$$VC^I(q) = 0.3q^2$$

How much would each of these plants want to supply at the price solved for in (d)?

Quantity: ____________________

g) Of course, we would expect their entry to lower prices. Solve for the new global supply curve and equilibrium price of aluminum.

Global Supply, $Q_S^G(p)$?: ____________________

Equilibrium Price: ____________

Which type of plant produces more aluminum – a traditional one, or an Icelandic one?

- ☐ Traditional
- ☐ Icelandic

2. *Picking up the Crumbs.* Crumbs Bakery has determined their daily total cost function (where quantity is in dozens of cupcakes) is

$$TC(q) = 225 + 6q + q^2$$

Unfortunately for Crumbs, cupcakes are commodities and there is free entry, making the market perfectly competitive.

(a) (3 points). What is Crumbs' supply function for dozens of cupcakes (for the range of prices where they are willing to supply any), **in terms of the market price for a dozen cupcakes**?

Supply function: $Q_S(P) =$ ____

(b) (3 points). If the current prevailing market price in the short run is \$4 per cupcake (so \$48 per dozen), how many cupcakes will Crumbs supply, and what will their daily profit be?

Supplied Quantity (dozens): $Q_S(P) =$ ____

Daily Profit: ____

(c) (3 points). Fads come and go, and the cupcake boom appears to be turning into a cupcake bust: market prices for cupcakes are falling. In the short-run, the fixed cost component of total costs is unavoidable. What is the price (per dozen cupcakes) below which Crumbs would stop baking entirely in the short term?

Price ____

(d) (4 points). In the long run, there is free entry, and indeed several competitors enter the cupcake market with the same cost function as Crumbs. What will be the long-run equilibrium price for a dozen cupcakes?

Price ____